

Representation of Signed binary Numbers

There are two methods to deal with signed binary numbers:

- (i) Sign-magnitude form
- (ii) Complement form

2's complement 2's complement

Sign magnitude form:

Sign-bit:

Left most bit of signed binary numbers is known as sign-bit.

$$= \begin{cases} 0 & \text{for +ive} \\ 1 & \text{for -ive} \end{cases}$$

for example:

+2	^{SB} 0 1 0	+6	^{SB} 0 0 1 0 0 0 0
-2	1 1 0	-6	1 1 0 1 0 0 0 0

In Sign magnitude form the left most bit represents the sign of binary number while remaining bits represent magnitude of binary number.

Syntax:

S.B	Actual binary
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e.g;

+2

S.B

0	1	0
---	---	---

→ magnitude bits

-2

1	1	0
---	---	---

1's Complement form:

⇒ In 1's complement form +ive numbers remains same as +ive magnitude numbers.

e.g

+25

$$= \begin{cases} \text{In sign-mag form} = (00010001)_2 \\ \text{In 1's comp form} = (00010001)_2 \end{cases}$$

⇒ In the 1's complement form of a -ive numbers the 1's complement of corresponding +ive number.

e.g.

$$-25 = \begin{cases} \text{In sign-magn form} = (10011001)_2 \\ \text{In 1's comp form} = (11100110)_2 \end{cases}$$

2's Complement form:

→ In 2's complement form of +ive numbers remains same as +ive sign-magnitude form.

e.g.

$$+25 = \begin{cases} \text{In sign-mag form} = (00011001)_2 \\ \text{In 2's comp form} = (00011001)_2 \end{cases}$$

→ In 2's complement form of -ive numbers is the 2's complement of corresponding +ive numbers.

e.g.

$$-25 = \begin{cases} \text{In sign-mag form} = (10011001)_2 \\ \text{In 2's comp form} = (111001101)_2 \end{cases}$$

Exmp: Represent -39 in Sign-magnitude and complement form. (8-bit)

$$-39 = (10100111)_2$$

2	39
2	19-1
2	9-1
2	4-1
2	2-0
2	1-0
	0-1

Sign magnitude form:

$$= (10100111)_2$$

1's comp form:

$$= (11011000)_2$$

2's comp form:

$$= (11011001)_2$$

Exp: Represent 29 in both form. (6-bit)

$$29 = (011101)_2$$

2	29
2	14-1
2	7-0
2	3-1
	1-1

Sign-mgn form:

$$(011101)_2$$

1's comp form:

$$= (011101)_2$$

2's compl form:

$$= (011101)_2$$

Exp: -99 represent in Sign-magnitude and complement form. (8-bits)

$$-99 = ?$$

2	99
2	49-1
2	24-1
2	12-0
2	6-0
2	3-0
	1-1

Sign-magnitude form:

$$= (10110001)_2$$

1's complement form:

$$= (11001110)_2$$

2's compl form:

$$= (11001101)_2$$

Addition of Signed Binary Numbers

Addition of signed binary numbers.

There are four cases:

→ Both numbers +ive

→ Both numbers -ive

→ Big +ive & little -ive

→ Little -ive big +ive

Case I : Both +ive

Exp: Add $(9)_{10}$ and $(6)_{10}$.

$$\begin{array}{r} +9 \\ +6 \\ \hline +15 \end{array}$$

$$\begin{array}{r} 2|9 \\ 2|4-1 \\ 2|2-0 \\ \hline 1-0 \end{array}$$

$$\begin{array}{r} 2|6 \\ 2|3-0 \\ \hline 1-1 \end{array}$$

$$\begin{array}{r} (1001)_2 \\ + (0110)_2 \\ \hline (1111)_2 \end{array}$$

Exp: Add $(28)_{10}$ and $(15)_{10}$.

$$\begin{array}{r} +28 \\ +15 \\ +43 \\ \hline \end{array} \quad = \quad \begin{array}{r} 00011100 \\ +00001111 \\ \hline 00101011 \end{array}$$

Decimal values of Signed Numbers

Case I: Sign-magnitude form:

Rule: Find the sum of weights of Magnitude bits then put (+ or -) according to the Sign bit.

Exp:

$$SB \leftarrow (10010101)_2 \Rightarrow (0010101)_2$$

$$\begin{array}{ccccccc} 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{array}$$

$$\downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow$$

$$0 + 0 + 16 + 0 + 4 + 0 + 1$$

$$= 21$$

$$= (-21)_{10}$$

$$\therefore \text{S.B is } 1$$

Exp:

$$(1100)_2 \Rightarrow (100)_2$$

$$1 \quad 0 \quad 0$$

$$4 + 0 + 0$$

$$= 4 \quad \{ \therefore \text{S.B is } 1 \}$$

$$= 4$$

Case II: 1's complement form

+ive Numbers
=> Simply find the sum of weights

Exp:

(00010111)₂

S.B → +ive

0 0 1 0 1 1 1

$$= 16 + 0 + 4 + 2 + 1$$

$$= +23$$

-ive Numbers
=> Find the sum of weights + 1

Exp:

(1110100)₂

S.B → -ive

1 1 1 0 1 0 0

$$= 64 + 32 + 16 + 4 + 0 + 0 + 0$$

$$= -78 + 1 = -77$$

Case III: 2's complement form

+ive numbers
=> Simply find the sum of weights

Exp:

(01010110)₂

0

1 0 1 0 1 1 0

↓ ↓ ↓ ↓ ↓ ↓ ↓

$$= 64 + 16 + 0 + 4 + 2 + 0 + 0$$

$$= +86$$

-ive numbers
=> Simply find the sum of weights

Exp:

(10101010)₂

1 0 1 0 1 0 1 0

↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓

$$= 128 + 0 + 32 + 0 + 8 + 0 + 2 + 0$$

$$= -170$$

∴ Note: same as 1's complement but don't add 1.

2's comp preferred

Case II: Both -ive

Exp: Add (-6) + (-13)

$$\begin{array}{r} 2 \mid 13 \\ 2 \mid 6-1 \\ 2 \mid 3-0 \\ \hline 1-1 \end{array}$$

$$\begin{array}{r} 2 \mid 6 \\ 2 \mid 3-0 \\ \hline 1-1 \end{array}$$

2's comp taking

$$+6 = 0110$$

$$\Rightarrow -6 =$$

$$1010$$

$$+13 = 1101$$

$$\Rightarrow -13 =$$

$$0001$$

$$= 111010$$

$$+ 110011$$

$$\boxed{1}10110!$$

Discard

Exp: Add (-32) + (-44)

$$-32$$

$$+32 = 100000$$

$$-44$$

$$+44 = 111100$$

$$\text{sum} = -76$$

$$-32 =$$

$$11100000$$

$$+ 11000100$$

Discard

$$\boxed{1}10100100$$

Final result

Case III: Big (+ive) and Little (-ive)

Exp: Add (28)₁₀ and (-15)₁₀

$$\begin{array}{r}
 +28 = 011100 \\
 +15 = 001111 \\
 -15 = \text{Taking 2's complement} \quad 110001
 \end{array}$$

$$\begin{array}{r}
 = 011100 \\
 + 110001 \\
 \hline
 \text{Discard } \boxed{1} 00101
 \end{array}$$

Case IV: Big (-ive) and Little (+ive)

Exp: Add (-28)₁₀ and (15)₁₀

$$\begin{array}{r}
 +28 = 011100 \\
 +15 = 001111 \\
 -28 = 100100
 \end{array}$$

$$\begin{array}{r}
 = 100100 \\
 + 001111 \\
 \hline
 \boxed{1} 10011
 \end{array}$$

Verification = Taking 2's complement

$$0110 / 01101$$

Subtraction of Signed binary numbers

Rule: Take 2's complement of Subtrahend and Add in Minuend.

Exp:

(a) $00001000 - 00000011$

Take 2's complement of Subtrahend.

$$\begin{array}{r}
 00001000 \rightarrow \text{Minuend} \\
 + 11111011 \rightarrow \text{Subtrahend} \\
 \hline
 \text{Discard carry } \boxed{1} 00000101 \rightarrow \text{Difference}
 \end{array}$$

(b) $00001100 - 11110111$

Taking 2's complement of Subtrahend.

$$\begin{array}{r}
 \text{Minuend} \leftarrow 00001100 \\
 \text{Subtrahend} \leftarrow + 00001001 \\
 \hline
 00010101 \rightarrow \text{Difference}
 \end{array}$$

(c) 1110011 - 00010011

Taking 2's complement of
Subtrahend

1110011

Taking 2's complement of
Subtrahend

$$\begin{array}{cccccccc}
 + & 1 & 1 & 1 & 0 & 0 & 1 & 1 \\
 & 1 & 1 & 0 & 1 & 1 & 0 & 1
 \end{array} \rightarrow \text{Minuend}$$

$\begin{array}{cccccccc} \boxed{1} & 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$

Discard
copy

(d) 1 0001000 - 11100010
Taking 2's complement of
Subtrahend.

Taking 2's complement of
Subtrahend.

Mirrored $\leftarrow$ 1 0 0 0 1 0 0 0
Subtracted $\leftarrow$ + 0 0 0 1 1 1 1 0

Subtrahend = $\begin{array}{r} 0001000 \\ + 0001110 \\ \hline \end{array}$

1 0 1 0 0 0 1 0 → Difference

Multiplication of Signed binary numbers

The numbers in multiplication are:

- ➔ The multiplicand
- ➔ The multiplier
- ➔ The product

There are two methods of multiplication:

Direct addition method

Direct addition method

- It is not efficient.
- It is time taking and long process method.

Exp: Multiply the signed numbers using direct method.

Multiplicand: 01001101

Multiplies: 00000100

$$\begin{array}{r} 01001101 \\ + 01001101 \\ \hline \end{array}$$

1 0 0 1 1 0 1 0

$$\begin{array}{cccccccc} + & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ \hline \end{array}$$

1 0 1 0 0 1 1 N

$\begin{array}{cccccccc} + & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ \hline 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$

Discard = 1 0 0 1 1 0 1 0 0

ii) Change any -ve number to its true form by taking its complement.

Find the partial products. Add the partial products to get final products.

(iii) if the sign of final product decided in:

(i) is -ve then take its 2's complement and attach a sign bit.

(ii) if it is +ve then leave its true form with 0 as a sign bit.

Example:- Multiply the signed binary numbers: 01010011 (multiplicand) and 11000101 (multiplier).

Take 2's complement of multiplier as it is -ve.

After taking 2's complement multiplier is 00111011

Now start multiply only magnitude bits:

$$\begin{array}{r} 1000101 \\ \times 0111011 \\ \hline 1000101 \\ 0001010 \\ 0010100 \\ 0001010 \\ 0000101 \\ 0000010 \\ 0000001 \end{array}$$

$$\begin{array}{r} 1010011 \\ \times 0111011 \\ \hline 1010011 \\ 1010011 \\ 1010011 \\ 1010011 \\ 1010011 \\ 1010011 \\ 1010011 \end{array}$$

↓ 2's complement taking
S.B 1011001101111
So, final product.

Multiply 0111111 by 00000101.

$$\begin{array}{cccccccc}
 & 1 & 1 & 1 & 1 & 1 & 1 & \\
 x & 0 & 0 & 0 & 0 & 1 & 0 & 1
 \end{array}$$

$$\begin{array}{cccccccc}
 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & x
 \end{array}$$

$$\begin{array}{cccccccc}
 0 & 1 & 1 & 1 & 1 & 1 & 1 & x \\
 1 & 1 & 1 & 1 & 1 & 1 & x & x
 \end{array}$$

$$\begin{array}{cccccccc}
 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & x
 \end{array}$$

$$\begin{array}{cccccccc}
 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & x
 \end{array}$$

$$\begin{array}{cccccccc}
 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & x & x
 \end{array}$$

$$\begin{array}{cccccccc}
 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & x & x
 \end{array}$$

$$\begin{array}{cccccccc}
 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & x & x
 \end{array}$$

$$\begin{array}{cccccccc}
 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1
 \end{array}
 \Rightarrow \text{Final}$$

1 1 0 0
1 0 1 1

Signed Division

Rules:-

- 1) Deciding the sign of Quotient
- 2) if the signs of divisor and dividend are same then quotient will be +ive.
- 3) if the signs of divisor and dividend are different then quotient will be -ive.
- 4) Initialize the quotient to zero.
- 5) Subtract the divisor from dividend using 2's C and find partial remainders. Also add 1 to the quotient.
- 6) if the partial remainder is zero or -ive the division is complete and if it is +ive then repeat step 3 until it becomes zero or -ive

Example:
Divide 01100100 by 00011001.

∴ Both numbers are +ve, the quotient will be +ve.

$$\begin{array}{r}
 01100100 \rightarrow \text{Div.} \\
 + 11001111 \rightarrow 2^3 \text{ c.} \\
 \hline
 1001001011 \rightarrow \text{Remainder} \\
 \text{Discard} \leftarrow 1 \\
 + 11001111 \\
 \hline
 1000010010 \\
 \text{Discard} \leftarrow 1 \\
 + 11001111 \\
 \hline
 1000011111 \\
 \text{Discard} \leftarrow 1 \\
 + 11001111 \\
 \hline
 1000000000 \rightarrow \text{Zero} \\
 \text{Discard carry}
 \end{array}$$

Add 1 to quotient.

$$\begin{array}{r}
 00000000 \\
 + 00000001 \\
 \hline
 00000001 \rightarrow \text{Again Add} \\
 + 00000001 \\
 \hline
 00000010 \rightarrow \text{Again Add} \\
 + 00000001 \\
 \hline
 00000111 \rightarrow \text{Again Add} \\
 + 00000001 \\
 \hline
 00001000 \rightarrow \text{Ans}
 \end{array}$$

Divide 00110000 by 00001100

Because divisor and dividend are +ve, so Quotient will be +ve.

$$\begin{array}{r}
 00110000 \\
 + 1100100 \text{ (23c of divisor)} \\
 \hline
 100100100 \\
 + 1110100 \\
 \hline
 100010000 \\
 + 1110100 \\
 \hline
 100000100 \\
 + 1110100 \\
 \hline
 100000000 \rightarrow \text{Zero remainder}
 \end{array}$$

Add 1 in Quotient in 4 times.

$$\begin{array}{r}
 00000000 \\
 + 00000001 \\
 \hline
 00000001 \\
 + 00000001 \\
 \hline
 00000010 \\
 + 00000001 \\
 \hline
 00000111 \\
 + 00000001 \\
 \hline
 00001000 \rightarrow \text{Final result}
 \end{array}$$